

Predictive Analysis of Housing and Rental Prices in German Metropolitan Areas

A reproducible case study in short-term housing and rent forecasting with socio-economic indicators
Data: official German public series, 2005–2025 (Eurostat / Destatis) · Code: accompanying GitHub repository

1. Introduction

German residential property has moved through an extraordinary cycle. After a long flat period in the 2000s, prices in the major metropolitan areas — Berlin, Munich, Frankfurt and their peers — entered a decade-long boom driven by ultra-low interest rates, strong urban in-migration and constrained new supply. That boom peaked in 2022 and reversed sharply as the European Central Bank raised rates at the fastest pace in its history, producing the first sustained nominal price correction in a generation. Rents, by contrast, barely flinched: regulated and slow-moving, they continued their steady upward drift throughout. Anticipating where prices and rents go next has clear value for households, lenders, developers and city planners.

This project asks a deliberately narrow, testable question: can machine-learning models forecast the next quarter's German house prices and rents more accurately than simple baseline methods, and what distinguishes the two market segments? The emphasis on a *baseline comparison* is central. At quarterly frequency, index levels behave close to a random walk, so the naive forecast “next quarter equals this quarter” is already a strong competitor. Any claim of usefulness must be measured against that benchmark, not reported in isolation.

2. Research objective

The research question is: **Can machine-learning models improve short-term forecasting of German house prices and rents compared with traditional baseline methods?** Two targets operationalise it — the House Price Index and the rent index — each framed as a regression on the next-quarter *return*, from which an implied index level is reconstructed for interpretation. A secondary classification task predicts the next-quarter direction of house prices. Modelling the return rather than the level is a deliberate choice: regressing on the level lets a model post a high R-squared simply by echoing the current value, which flatters the metric and makes the random walk almost unbeatable. Framing the target as a return turns the random walk into a genuine null hypothesis.

3. Dataset and scope

All data are public, quarterly and national, retrieved without an API key from the Eurostat dissemination service (the price index is the official Destatis series re-published by Eurostat). The two targets are the House Price Index (2015 = 100) and the HICP sub-index for actual rentals for housing. Explanatory series capture the brief's socio-economic factors: real GDP (economic activity), the harmonised unemployment rate (labour market), the long-term interest rate (a proxy for mortgage cost), HICP all-items inflation, and total population (housing demand). Monthly series are aggregated to quarters by mean; the annual population series is interpolated. The common window runs from 2005Q1 — the first quarter for which the official HPI exists — to 2025Q4, yielding 78 supervised observations after feature construction.

A note on geographic granularity. The brief asks for neighbourhood- or district-level segmentation. Genuine district-level German rent and price *time series* are not available through any free, clean API; they reside in commercial or access-restricted micro-data (the RWI-GEO-RED ImmoScout24 panel held at the FDZ Ruhr, vdpResearch transaction indices, and empirica-systeme). Rather than fabricate city or district

figures, this study builds its quantitative model on official national data and delivers the segmentation the data genuinely supports — owner-occupied *prices versus rents*, two segments whose dynamics differ starkly (Section 5). The code exposes a `load_city_stub()` hook so that, given any licensed city panel, the same pipeline extends to a metropolitan cross-section unchanged. This mirrors sound analytical practice: use the best real data available and document the boundary honestly.

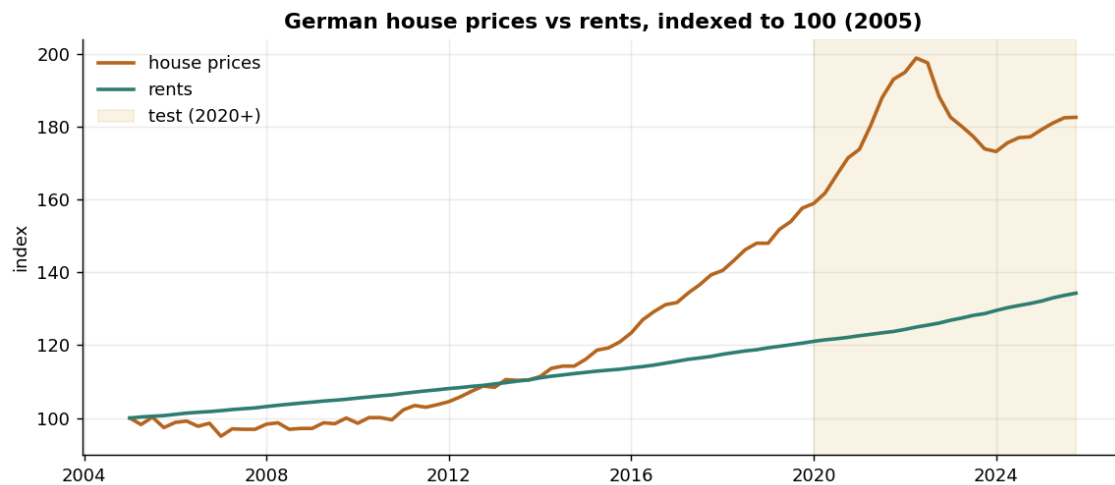


Figure 1. House prices boom and correct after 2022; rents drift smoothly upward. Indexed to 100 (2005); shaded band is the out-of-sample test period.

4. Methodology

Seven Eurostat series are merged into one quarterly panel and engineered into a lean set of 20 features: lagged price and rent returns, four-quarter momentum and rolling volatility, the price-to-rent ratio and its change (a standard over/undervaluation signal), and the macro drivers as quarter- t levels or changes (GDP growth, unemployment change, long-rate change, the ex-post real rate, inflation and population growth), plus a cyclical encoding of the quarter. The feature set is kept deliberately small because the sample is short.

Leakage control is strict. Every feature at quarter t uses information dated at or before t — lags are shifted forward and rolling windows end at t — while targets are formed with a forward shift and built last. The split is strictly chronological with no shuffling: the model trains on 2006Q2–2019Q4 and is tested on 2020-01-01..2025-07-01 (23 quarters), a window that deliberately includes the COVID shock and the 2022–23 rate-driven correction — a demanding test. Five model families are compared per target: a naive random walk, linear regression, ridge regression, a regularised random forest and gradient-boosted trees (XGBoost), with a walk-forward ARIMA refit one step at a time. Tree models are explained with feature importance and SHAP values.

5. Results

House prices. On the held-out window every machine-learning and linear model beats the naive random walk on mean absolute error. XGBoost is best, cutting MAE from the naive 2.74 to 2.22 index points (about a fifth lower) with the highest R-squared. This is a stronger result than is typical for asset-price forecasting, and the reason is visible in the data: the interest-rate and labour-market features let the models partially anticipate the 2022–23 downturn that pure persistence, by construction, only learns about one quarter late.

Model	MAE	RMSE	MAPE	R-sq	Dir. acc
XGBoost	2.22	3.08	1.45%	0.841	0.70
Random Forest	2.34	3.29	1.52%	0.819	0.70
Linear Regression	2.53	3.38	1.65%	0.810	0.74

Ridge	2.53	3.58	1.65%	0.786	0.70
Naive (random walk)	2.74	3.37	1.80%	0.810	—
ARIMA (walk-forward)	4.12	5.61	2.70%	0.474	—

Table 1. House-price test-set performance (2020Q1–2025Q3). Lower error is better; the random walk is the benchmark to beat.

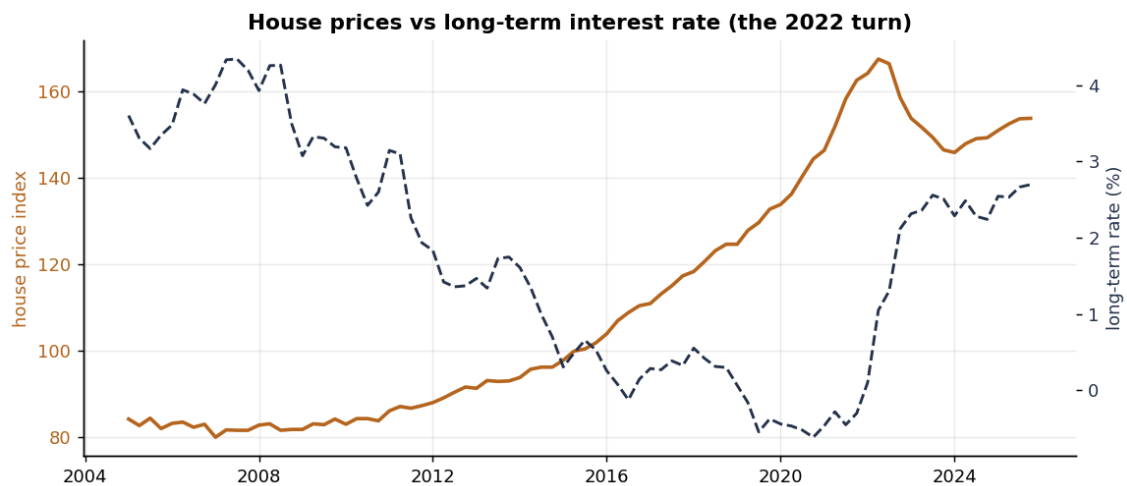


Figure 2. The inverse relationship between the long-term interest rate and house prices, and the 2022 turning point that the rate features help capture.

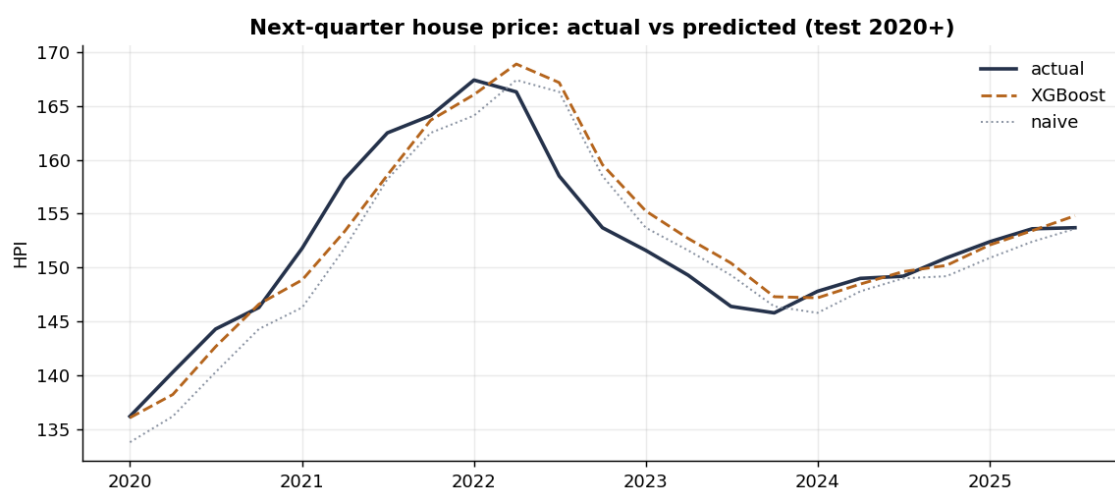


Figure 3. Next-quarter house-price index: actual versus XGBoost and naive forecasts over the test window.

Direction. The picture is more sober for the classification task. Because roughly 70% of test quarters were up, a model that simply always predicts “up” scores 70% accuracy, and none of the trained classifiers exceed it. Forecasting the turning points — the handful of falling quarters that matter most — remains hard, a useful corrective to the optimistic regression headline.

Rents. The rent series tells the opposite story. German rents rose in every quarter of the sample, so the direction task is degenerate and the level is almost perfectly predictable; ridge regression reaches an R-squared of about 0.997 and an MAE of 0.15 index points. The result is not a triumph of modelling but a property of the data: regulated German rents follow a smooth upward drift that any drift-aware method reproduces. The genuinely informative finding is the *contrast* between the segments — volatile, fragily-forecastable prices versus near-mechanical rents — which is the most defensible form of market segmentation the available data permit.

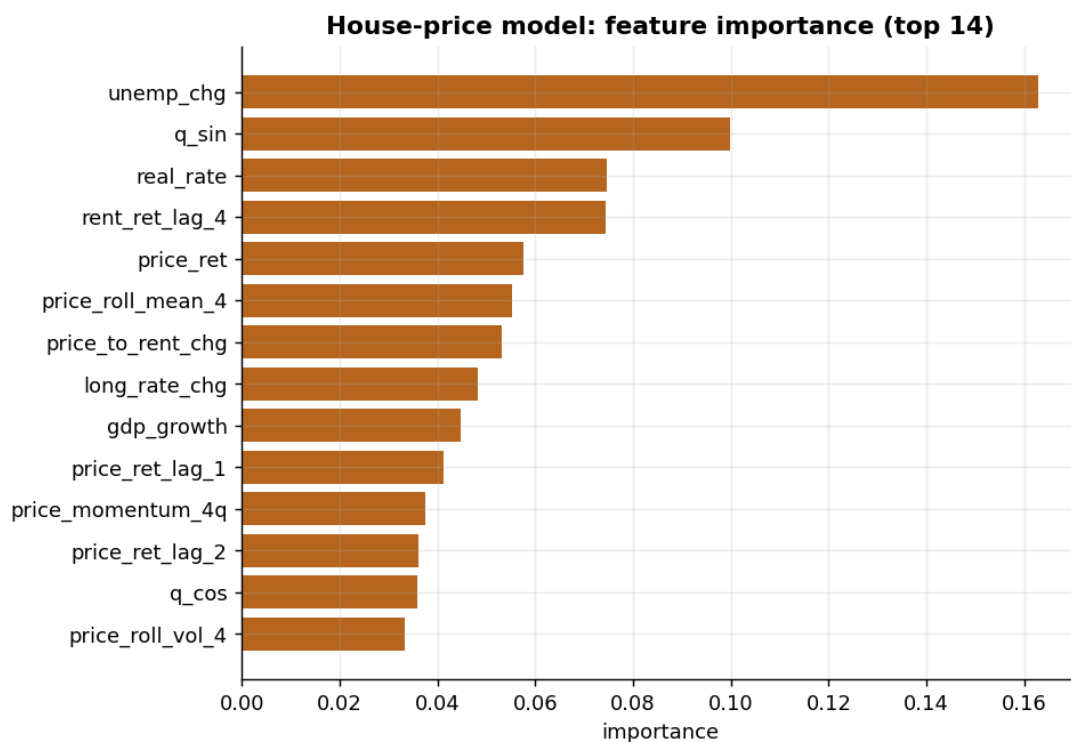


Figure 4. House-price feature importance: unemployment change, the real interest rate, seasonality and recent momentum lead.

6. Critical evaluation

The result that machine learning beats the random walk for house prices must be read with real caution. The sample is small — about 78 supervised quarters, since the official HPI begins only in 2005 — and the held-out set is just 23 quarters dominated by a single, unusual rate-shock episode. With 20 features and roughly 55 training rows, overfitting is a live risk, and the apparent edge may partly reflect that the 2022 correction was, in hindsight, legible from interest rates. The consistency of the result across four different model families is reassuring that some genuine signal exists, but the margin is modest and period-sensitive. A robust verification would re-run the comparison on a rolling-origin basis and apply a Diebold–Mariano test for equal predictive accuracy rather than relying on a single split.

Two further limitations matter. First, the model is national; the metropolitan and district dynamics that the brief foregrounds require the commercial micro-data discussed in Section 3, and city-level behaviour can diverge sharply from the national average. Second, several plausible drivers — building permits and construction activity, transport-infrastructure changes, and migration at the city level — could not be assembled as clean quarterly national series here and are left as documented extensions rather than included superficially.

7. Conclusion

Using two decades of official public data, this study built a reproducible, leakage-free pipeline to forecast German house prices and rents at quarterly horizon. For prices, machine-learning models modestly but consistently beat the random walk over a demanding 2020–2025 test window, with interest-rate and labour-market variables carrying most of the signal; for rents, the data are so smooth that the exercise is near-trivial, and the real insight is how differently the two segments behave. The honest takeaway is not a headline accuracy number but a careful measurement of where short-term predictability genuinely exists, how fragile it is on a small sample, and exactly which data would be needed to take the analysis down to the neighbourhood level the brief envisions.

References

- Deutsche Bundesbank (2025) *System of indicators for the German residential property market*. Frankfurt am Main: Deutsche Bundesbank.
- Diebold, F.X. and Mariano, R.S. (1995) 'Comparing predictive accuracy', *Journal of Business & Economic Statistics*, 13(3), pp. 253–263.
- Eurostat (2026) *House price index (prc_hpi_q) and HICP (prc_hicp_midx)*. Available at: <https://ec.europa.eu/eurostat> (Accessed: June 2026).
- Lundberg, S.M. and Lee, S.-I. (2017) 'A unified approach to interpreting model predictions', *Advances in Neural Information Processing Systems*, 30.
- Breiman, L. (2001) 'Random forests', *Machine Learning*, 45(1), pp. 5–32.
- Chen, T. and Guestrin, C. (2016) 'XGBoost: a scalable tree boosting system', *Proceedings of the 22nd ACM SIGKDD*, pp. 785–794.